

A Comprehensive Survey of Knowledge Graph Reasoning: Approaches and Applications

Guanglin Niu, Bo Li, Yangguang Lin

 <https://github.com/ngl567/KGR-Survey>

Abstract—Knowledge graph reasoning (KGR) aims to infer novel knowledge based on existing facts in knowledge graphs (KGs), playing a crucial role in various cognition intelligence systems across diverse domains. The previous review works explore KGR models from specific perspectives such as KG types and embedding spaces. In contrast, this survey provides a more comprehensive perspective of KGR from foundational approaches and their applications. Notably, some seldom-attended approaches such as negative sampling strategies, popular open-source libraries, and rule-guided KGR paradigms are carefully reviewed. Besides, we explore advanced techniques, such as large language models (LLMs) and their impact on KGR. The comparison among foundational models are analyzed to declare their strengths and limitations. More interestingly, this is the first effort to provide a taxonomy of real-world KGR applications for both horizontal and vertical domains. Furthermore, we highlight the challenges and opportunities in the field of KGR, including trustworthiness, multimodal reasoning, continual learning, uncertainty and LLM-driven approaches. This work aims to bridge the gap between theoretical advancements and practical deployment of KGR models, and outline promising future directions.

Index Terms—Knowledge graph reasoning, multi-hop and temporal reasoning, knowledge graph embedding, downstream application, large language model.

I. INTRODUCTION

Deep learning has achieved remarkable success in perceptual intelligence tasks, capitalizing on substantial computational power and vast quantities of labeled data. However, its limitations in knowledge representation hinder its ability to perform complex tasks such as knowledge reasoning. In contrast, knowledge graphs (KGs) have emerged as a powerful technique for modeling and utilizing knowledge by representing entities and their relations through a directed graph structure [1], [2], including notable examples such as Freebase [3], YAGO [4], WordNet [5], NELL [6], and DBpedia [7].

In real-world scenarios, most KGs are inevitably incomplete. For instance, 75% of the people in Freebase lack nationality information [8], and 60% of the people in DBpedia lack birthplace information [9]. Thus, knowledge graph reasoning (KGR) (also known as knowledge graph completion) techniques [10] aim to discover semantic associations from the existing knowledge in a KG and infer unknown facts. At present, KGR techniques have been widely applied in various

practical scenarios, including drug discovery [11], bond market risk supervision [12], product recommendation [13], voice assistants [14], and intelligent education [15].

For downstream applications, selecting an appropriate KGR paradigm primarily depends on the structural complexity of queries, the need for path-level interpretability, and the temporal dynamics of facts. (1) Traditional (single-step) KGR suits triple completion and sparsity mitigation such as product recommendation, biomedical fact augmentation [11], [12], where missing entities in static Triples are directly predicted by KG embedding (KGE) or logic rules. (2) Multi-hop KGR is required when answers emerge only through compositional constraints, as in complex KGQA [16] or explainable recommendation that justifies user-item relations via semantic paths [17]. (3) Temporal KGR contributes to time-aware QA and event/risk forecasting, where validity intervals and historical evolution affect inference [18].

In recent years, several studies such as [19]–[23] have reviewed the KGR techniques from various perspectives. For instance, [19] summarizes the KGR models according to KG types, [20] and [21] review the multi-hop KGR approaches for traditional KGs, while [22] and [23] analyze the existing temporal KGR models. However, the previous review works mainly introduce the KGR models specific to KG types or reasoning scenarios, lacking a systematic summarization regarding both the theoretical foundations and the practical applications. Besides, these reviews typically present KGR applications from a single perspective, either horizontal or vertical. In contrast, this survey is structured around downstream applications, systematically summarizing and critically evaluating theories and practical methodologies in KGR.

Accordingly, this survey first summarizes the foundational paradigms of traditional KGR (Section III), covering purely triple-based KGE models and their variants, as well as some representative open-source libraries. We then review logic-based KGR, including inductive logic programming, neural rule learning, and joint rule-KGE frameworks. Subsequently, we summarize multi-hop and temporal KGR (Section IV), which form the methodological core of many downstream application scenarios. Significantly, it is the first effort to categorize and introduce KGR applications spanning both horizontal domains namely general-purpose application categories and vertical domains namely domain-specific settings (Section V). Finally, this survey discusses several challenges that require to be addressed for the practical application of KGR technology, highlighting the research hotspots and future directions worth further exploring (Section VI).

Guanglin Niu and Bo Li are with the School of Artificial Intelligence, Beihang University. Yangguang Lin is with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications. (E-mail: beihangngl@buaa.edu.cn, boli@buaa.edu.cn, sunshinelin@bupt.edu.cn.)

Corresponding author: Bo Li. This work was partially supported by the National Natural Science Foundation of China (No. 62376016, U25A20531).

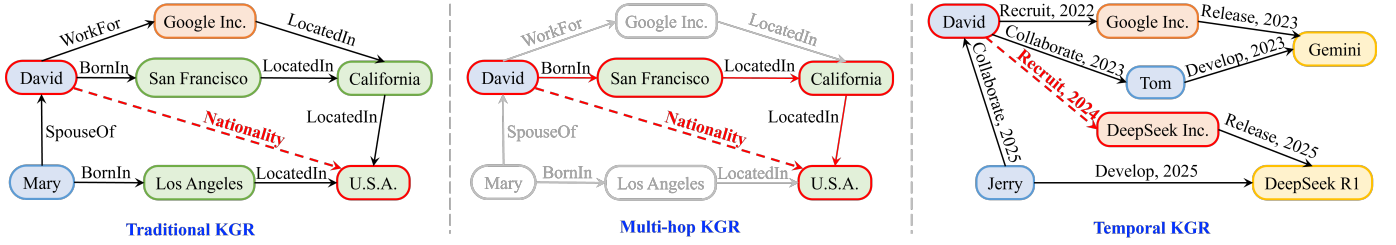


Fig. 1. Overview of three KGR paradigms and their underlying query structures: traditional KGR performs single-step reasoning over triples, multi-hop KGR answers queries by traversing paths from head to tail entities, temporal KGR completes time-stamped events represented as quadruples.

II. PRELIMINARY

A. Knowledge Graphs

Knowledge graphs (KGs) organize heterogeneous information through entities and relations, with notable examples such as Freebase [3] and YAGO [4]. Formally, a KG can be represented as $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$, where $\mathcal{E}$, $\mathcal{R}$ and $\mathcal{T}$ denote entity set, relation set and triple set, respectively. Particularly, $(h, r, t) \in \mathcal{T}$ indicates each triple in the KG, in which $h \in \mathcal{E}$, $r \in \mathcal{R}$ and $t \in \mathcal{E}$ denote head entity, relation and tail entity.

Temporal KGs extend traditional KGs with a time dimension $\mathcal{T}_{time}$ such as GDELT [24], where events are valid within specific time intervals or timestamps. Each event in a TKG is represented as $\mathcal{G}_t = (\mathcal{E}, \mathcal{P}, \mathcal{T}_{time}, \mathcal{Q})$, in which $\mathcal{E}$, $\mathcal{P}$ and $\mathcal{Q}$ denote the entity set, predicate (namely relation) set and quadruple set, respectively. Each quadruple indicates an event in the form of $(s, p, o, time)$ where $s \in \mathcal{E}$, $p \in \mathcal{P}$, $o \in \mathcal{E}$, and $time \in \mathcal{T}_{time}$ are subject, predicate, object and timestamp/time interval.

B. Knowledge Graph Reasoning

In this paper, we focus on three most common KGR tasks in downstream applications as shown in Fig. 1. Traditional KGR [21] performs single-step triple completion on static graphs, multi-hop KGR [25] conducts multi-step path-based reasoning to exploit intermediate relational semantics, and temporal KGR [26] extends reasoning to time-stamped events by modeling temporal validity and evolution. For an instance specific to traditional KGR, given the head entity *David* and the relation *Nationality*, the tail entity *U.S.A.* should be predicted as the result without any intermediate reasoning steps or time constraints. In contrast, in terms of multi-hop KGR, predicting the tail entity with the given head entity *David* and relation *Nationality* involves analyzing multi-hop paths in the KG namely *David* $\xrightarrow{\text{BornIn}}$ *San Francisco* $\xrightarrow{\text{LocatedIn}}$ *California* $\xrightarrow{\text{LocatedIn}}$ *U.S.A.*. Besides, given a query entity *David* and query predicate *Recruit* with the specific timestamp year 2024, temporal KGR aims to predict the event $(\text{David}, \text{Recruit}, ?, 2024)$.

C. Performance Evaluation Metrics

For all KGR tasks, the objective is to recover the missing element of a factual triple or event quadruple. Evaluation follows a standard ranking protocol. For head entity prediction $(?, r, t)$, every entity in the KG is substituted into the head position to

form candidate triples. The model scores all candidates and the rank of the true triple is recorded. Tail entity prediction is performed analogously. Thus, a rank is obtained for each gold answer. Three widely used metrics [27] are then reported:

(1) MR: The mean rank of all correct triples, calculated as: $MR = \frac{1}{N} \sum_{i=1}^N rank_i$, where $rank_i$ represents the rank of the correct triple corresponding to the i^{th} test sample, and N is the total number of test samples. (2) MRR: The mean reciprocal rank of all correct triples, calculated as: $MRR = \frac{1}{N} \sum_{i=1}^N \frac{1}{rank_i}$. (3) Hits@n: The proportion of correct triples ranked within the top n , calculated as: $Hits@n = \frac{1}{N} \sum_{i=1}^N I(rank_i \leq n)$, where $I(rank_i \leq n)$ is an indicator function that equals 1 if $rank_i \leq n$ is true. Typically, the value of n is always set as 1, 3 or 10.

Notably, a lower MR, a higher MRR, and a higher Hits@n indicate superior performance of KGR models. MR is sensitive to outliers, potentially compromising its robustness in assessing overall reasoning results. In contrast, MRR effectively mitigates this issue, making MRR and Hits@n more reliable and universal evaluation metrics. However, the presence of other correct candidates ranked before the correct prediction result affect the reliability of reasoning results. Therefore, existing strategies filter reasoning results by removing candidate triples already existing in the KG.

III. TRADITIONAL KGR

For traditional KGR, existing studies can be classified into KGE models and logic-based models. KGE models learn entity and relation representations as vectors or matrices, and then score candidate triples that satisfy the constraints defined in KGE models as inference results. Besides, logic-based models focus on learning logic rules from the KG to infer triples from observed triples and matched rules. For instance, the observed triples $(\text{David}, \text{BornIn}, \text{San Francisco})$ and $(\text{San Francisco}, \text{LocatedIn}, \text{U.S.A.})$ in Fig. 1 could match the logic rule $\text{Nationality}(x, y) \Leftarrow \text{BornIn}(x, z) \wedge \text{LocatedIn}(z, y)$. Consequently, the rule is instantiated as $\text{Nationality}(\text{David}, \text{U.S.A.}) \Leftarrow \text{BornIn}(\text{David}, \text{San Francisco}) \wedge \text{LocatedIn}(\text{San Francisco}, \text{U.S.A.})$ to achieve the predicted result $(\text{David}, \text{Nationality}, \text{U.S.A.})$.

A. KGE Models

KGE technique is the most popular mechanism to evaluate the plausibility of each triple via calculating the score derived from the pre-defined scoring function and KG embeddings.

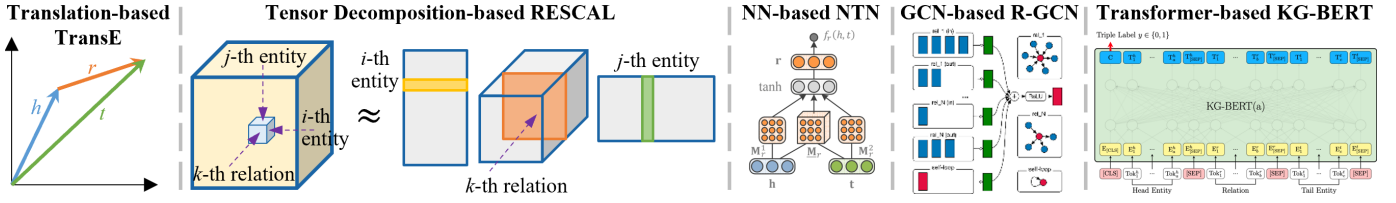


Fig. 2. The illustration of five representative KGE models. The diagrams of NTN [28], R-GCN [29] and KG-BERT [30] are derived from original papers.

This section reviews and introduces various KGE models that (1) rely solely on factual triples, (2) incorporate ontology information, (3) integrate path information, (4) leverage multimodal information and (5) negative sampling strategies for training KGE models as well as (6) open-source libraries of implementing KGE models.

1) **KGE Model with Triples:** KGE models differ primarily in how they represent a triple in terms of the interaction among entity pairs and their linked relations. They can be grouped into five categories: (a) translation-based models, (b) tensor decomposition-based models, (c) neural networks-based models, (d) graph neural networks-based models, and (e) Transformer-based models, each exemplified in Fig. 2. All models share three design choices: the embedding space, the encoding of entities/relations, and the scoring function that measures triple plausibility.

(a) **Translation-based Model:** Inspired by the word embedding method word2vec [31], TransE [32] represents relations as translation operations in the embedding space, as illustrated in Fig. 2. For a correct triple (h, r, t) , the optimization objective can be formulated as:

$$\mathbf{h} + \mathbf{r} \approx \mathbf{t} \quad (1)$$

where $\mathbf{h}$, $\mathbf{r}$ and $\mathbf{t}$ denote the embeddings of head entity h , relation r , and tail entity t in a real-valued Euclidean space. The corresponding scoring function is defined as:

$$E(h, r, t) = \|\mathbf{h} + \mathbf{r} - \mathbf{t}\| \quad (2)$$

where $\|\cdot\|$ represents either the $L1$ or $L2$ norm.

TransH [33] and TransR [34] extend TransE by giving each relation its own projection (a hyperplane or relation space) so entities acquire relation-specific representations, alleviating one-to-many and many-to-many issues. Their projections, however, depend only on the relation and ignore asymmetric semantic roles of head and tail. TransD [35] addresses this with distinct dynamic projection matrices for head and tail. TransSparse [36] adapts matrix sparsity to relation-entity connectivity. PairE [37] learns pairwise relation embeddings to project head and tail separately. TransA [38] employs a symmetric non-negative relation matrix with an adaptive Mahalanobis distance.

KG2E [39] introduces uncertainty by modeling entities and relations as Gaussian distributions (means encode position, covariances encode uncertainty). ManifoldE [40] constrains the tail to a hypersphere manifold centered at the head with radius defined by the relation. TorusE [41] embeds on a compact Lie group under a TransE-style objective. Hierarchical structure can be captured in hyperbolic space: Poincaré [42]

maps more general concepts near the origin and specific ones toward the boundary. MuRP [43] adds Möbius matrix-vector operations for relation-aware projections. Alternatively, polar decomposition encodes hierarchy: HAKE [44] uses modulus for level and phase for intra-level distinction, while H2E [45] and HBE [46] fuse hyperbolic geometry with polar coordinates for richer hierarchy modeling.

RotatE [47] models relations as element-wise complex rotations (Hadamard product), naturally handling symmetry via $\mathbf{r} \circ \mathbf{r} = 1$, which pure translations cannot. QuatE [48] generalizes rotation to quaternion space through Hamilton product, enabling higher-dimensional rotational interactions. DualE [49] further unifies translation and rotation in dual quaternion space, increasing expressiveness over QuatE.

(b) **Tensor Decomposition-based Model:** RESCAL [12] regards the KG as a third-order tensor and decomposes each relation into a full matrix. DistMult [50] restricts the relation embeddings to diagonal matrices for efficiency. ComplEx [51] uses complex-valued vectors and the Hermitian product while HolE [52] employs circular correlation, enabling to model both symmetry and anti-symmetry relations. TuckER [53] exploits Tucker decomposition for KGE, in which explicit core tensor models relation interactions. MIG-TF [54] augments Tucker factorization with a hyperbolic interaction correction to capture hierarchical distributions.

(c) **Neural Network-based Model:** SME [55] is an early neural networks (NNs)-based KGE that embeds entities and relations, combines (h, r) and (t, r) in a hidden layer, and scores a triple via the dot product of the two hidden states. NTN [28] replaces this with relation-specific bilinear/tensor transformations followed by a nonlinearity. NAM [56] concatenates the head and relation, feeds them to a network, and fuses the output with the tail representation to produce the score. ConvE [57] and ConvKB [58] reshape embeddings into 2D and apply convolutions to model interactions. SPR [59] studies the effects of normalization and regularization for NN-based KGE on sparse KGs.

(d) **Graph Neural Network-based Model:** GNNs naturally suit KGs by aggregating relation-aware neighbors to learn contextualized entity embeddings. R-GCN [29] performs relation-specific message passing to encode local structure. SACN [60] augments R-GCN with weighted convolution and matrix-based entity/relation embeddings, and decodes via ConvE-style convolutions with a TransE-like score. KBGAT [61] applies multi-head attention to emphasize informative neighbors. KE-GCN [62] iteratively refines embeddings by coupling graph convolution with KGE scoring. NoGE [63] further incorporates co-occurrence patterns into GNN aggregation.

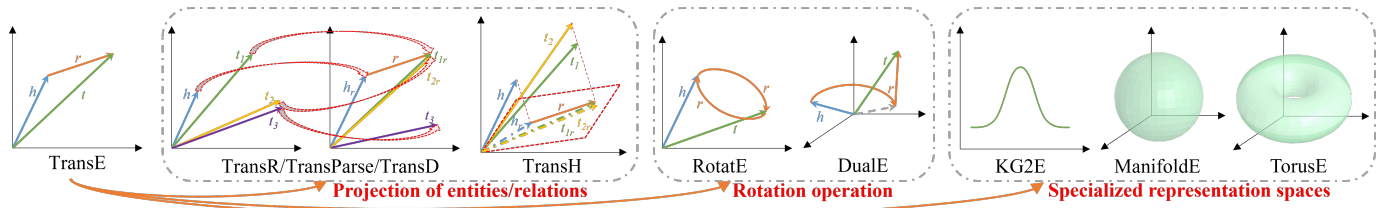


Fig. 3. The illustration of TransE and three categories of its variants.

Besides, graph learning on KGR has also been explored for unsupervised tasks such as multi-view clustering [64].

(e) **Transformer-based Model:** Transformers, successful in NLP and CV [65], have been adapted to KGE for modeling long-range dependencies and contextual interactions. R-MeN [66] uses self-attention in a relation memory network to capture latent triple-memory dependencies. CoKE [67] and HittER [68] cast KGR as a cloze task over triples. GenKGC [69] formulates KGR as sequence-to-sequence with pre-trained language models, while iHT [70] pre-trains on large KGs for transferable representations. Relphormer [71] tailors attention to KG topology via dynamically sampled subgraph sequences. KoPA [72] bridges KGE and LLMs with knowledge prefix adapters. KICGPT [73] combines an LLM with triple retrieval and in-context learning to mitigate data sparsity without additional training.

2) **Ontology-Aware KGE Model:** Traditional KGE models usually rely only on factual triples, whereas real-world KGs are often built together with an ontology that encodes hierarchical concepts and inter-concept relations, such as *U.S.A.* as a *Country* and *(Person, Nationality, Country)* at the concept level. To leverage such high-level semantics, ontology-aware KGE techniques fall into two lines: ontology graph-based and entity type-based models.

(a) **Ontology Graph-based Model:** A simple approach augments the instance graph with instance-of links and trains on the enlarged triple set [74], but this may conflate entities and concepts. JOIE [75] jointly embeds ontology and instance layers by exploiting concept hierarchies. CISS [76] models structural similarity and inheritance via class sets and a virtual entity layer, aligning subgraph structures across layers. CAKE [77] builds a commonsense graph unifying ontology and instances and performs coarse-to-fine two-view reasoning. Wang et al. [78] inject ontology constraints (hierarchies, attributes) directly into embedding learning. Concept2Box [79] uses box embeddings to capture overlapping and hierarchical concept relations.

(b) **Entity Type-based Model:** Here, concepts act as types guiding entity representations. SSE [80] clusters entities by type. Krompaß et al. [9] exploit head/tail type priors to improve negative sampling. Since entities are often multi-typed, TKRL [81] learns type embeddings at multiple levels, and TransET [82] projects entities into their type spaces. These use static types. AutoETER [83] instead generates contextualized type embeddings per triple by combining factual and commonsense semantics.

Note that ontology and instance graphs usually carry different relation sets: ontology primarily enriches entity se-

mantics rather than directly enabling inference. Moreover, many KGs lack complete type annotations (NELL's coarse domains, WordNet's missing explicit types), which limits the applicability of ontology-driven models.

3) **Path-Enhanced KGE Model:** The path-enhanced KGE models can be classified into the following three types according to the characteristics of paths.

(a) **Relational Path-based Model:** The existing approaches like Path-RNN [84], PTransE [85], and PRN [86] integrate the relations along a path using RNNs or algebraic operations namely addition and multiplication, embedding the path into TransE-based models. To capture the order of relations, OP-TransE [87] projects the head and tail entities into separate spaces and introduces a sequence matrix. To account for varied semantic importance among path relations, HARPA [88] applies hierarchical attention to focus on informative relations. RPJE [89] learns the path embeddings by incorporating symbolic rules for transparent path composition, improving both explainability and reasoning accuracy.

(b) **Complete Path-based Model:** The aforementioned relational path models often ignore entities, complete path-based methods integrate both entities and relations. PARL [90] models paths as entity-relation sequences using RNNs. However, the semantic heterogeneity between entities and relations would limit the performance. To address this, RPE [91] maps entities into both relation and path spaces to reduce the heterogeneity issue. Das et al. [92] and Jiang et al. [93] average entity type embeddings and combine these with relation embeddings to improve path representation.

(c) **Multiple Path-based Model:** Recognizing that multiple paths may connect an entity pair, multiple path-based models aim to identify and integrate the most informative ones. CPCovKE [94] uses gated mechanisms to filter noisy paths. PaSKoGE [95] models the interaction between paths and relations, dynamically adjusting the margin in the loss function. PathCon [96] introduces relation message passing and attention-based aggregation to encode path semantics. PTrustE [97] constructs a path reliability network and uses bi-directional gated units to compute a path reliability score. TAPR [98] uses attention to assign weights to paths, fusing the results for final prediction. Niu et al. [99] combine rule-driven and data-driven strategies, embedding each path and reasoning via a bi-directional mechanism.

4) **Multimodal-Enhanced KGE Model:** : In real-world KGs, entities often have rich multimodal information such as images, text descriptions, and numerical attributes. To leverage this auxiliary data, multimodal-enhanced KGE (MKGE) models have been developed. In terms of the scheme of learning

and applying multimodal embeddings, there are three streams for MKGE tasks: (a) text-based MKGE model, (b) fusion-based model and (c) integration-based model.

(a) **Text-based MKGE Model:** These methods exploit only textual descriptions (names, anchors, definitions) to enrich structural embeddings. Wang et al. [100] first align entities with their descriptions for unified representations. DKRL [101] encodes descriptions via bag-of-words and CNN and combines them with a TransE-style objective, outperforming TransE [32]. TEKE [102] refines description usage through entity linking and contextual word extraction. KG-BERT [30] directly models triples as text sequences with a BERT encoder and treats KGR as triple classification. SimKGC [103] applies contrastive learning over PLM-encoded descriptions to sharpen discrimination. StAR [104] further preserves graph topology while leveraging PLM-derived textual semantics. RAA-KGC [105] explores relation-aware anchor enhancement for improving completion performance by incorporating anchor-based signals in a relation-dependent manner.

(b) **Fusion-based Model:** Fusion approaches jointly encode multiple modalities (graph, text, images) to obtain complementary entity representations. IKRL [106] projects image features into the entity space for TransE-style interaction. TransAE [107] and RSME [108] integrate visual and structural signals. RSME adds gating to suppress noisy images. Cao et al. [109] employ optimal transport (Wasserstein alignment) for distribution-level fusion. Hybrid Transformer [110] uses multi-level fusion to combine multimodal evidence for knowledge graph completion. MGKsite [111] builds a KNN graph and applies a GNN to correlate visual, textual, and acoustic cues. MKBE [112] treats modality content as auxiliary triads via combinatorial coding. NativE [113] introduces relation-guided bi-adaptive fusion plus adversarial augmentation to mitigate modality imbalance. MMKGR [114] uses gated attention for noise suppression and cross-modal complementarity. While efficient and scalable, fusion quality can still degrade when a modality is sparse or noisy.

(c) **Integration-based Model:** Integration-based (ensemble) models train modality-specific components separately and fuse their outputs. MoSE [115] decouples tightly entangled multimodal signals by independently training three KGR models on factual triples, text, and images, then adaptively re-weighting modalities at inference. IMF [116] adopts distinct encoders for structural, visual, and textual features and aggregates their scores to form the final prediction. This strategy fully leverages complementary modality information and maintains robustness when one modality underperforms, but increases computational cost due to multiple model trainings.

5) **Negative Sampling for KGE:** KGR follows the open-world assumption [117], where missing triples may be either false or simply unobserved. Since KGs rarely contain explicit negatives, KGE models with pairwise losses have to synthesize negatives. The standard practice corrupts a positive triple by replacing an entity or relation under the local closed-world assumption [118]. The quality and efficiency of negative sampling strongly affect training stability and performance [119].

As shown in Fig. 4, negative sampling strategies for KGE can be grouped into six categories:

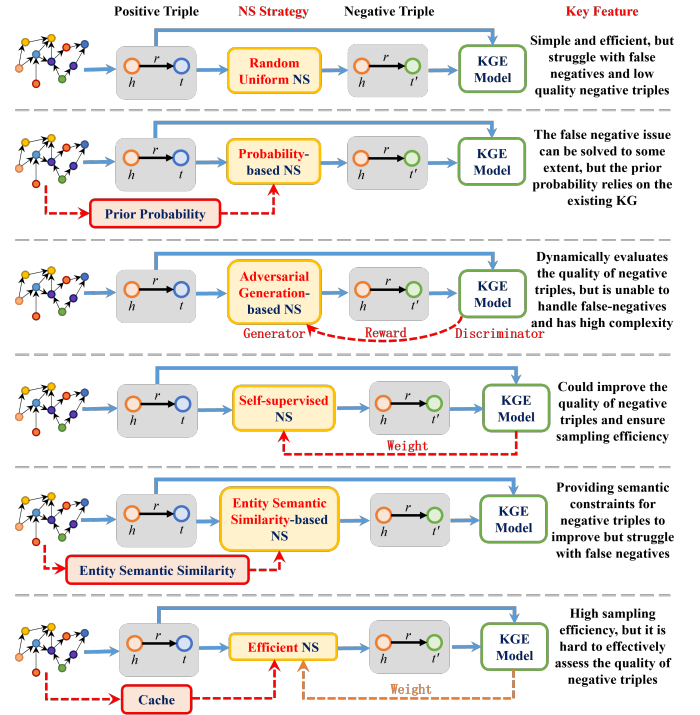


Fig. 4. The illustration of six types of negative sampling (NS) strategies.

(a) **Random Uniform Negative Sampling:** Randomly replace head entity, tail entity, or relation with a uniform distribution [33]. Batch NS [120] improves throughput at the batch level. This simple scheme often yields overly easy negatives implying weak gradients in training or false negatives namely true but unobserved triples, harming optimization.

(b) **Probability-based Negative Sampling:** Bernoulli sampling [121] adapts replacement probabilities to relation arity. Zhang et al. [122] extend this to relation replacement. SparseNSG [123] leverages relation-entity co-occurrence and false-negative estimation. These methods reduce false negatives over uniform sampling but heavily depend on KG statistics and are less adaptive.

(c) **GAN-based Negative Sampling:** KBGAN [124] employs a generator-discriminator (GAN) [125] to propose hard negatives. IGAN [126], GraphGAN [127], KSGAN [128], and RUGA [129] refine generation and selection. FedEAN [130] introduces entity-aware adaptive GAN sampling. GAN-based schemes yield informative negatives and adaptive distributions but risk false negatives and add notable complexity.

(d) **Self-supervised Negative Sampling:** Self-Adv [47] avoids an external generator by weighting negatives via the model's own scores (self-adversarial). ASA [131] emphasizes medium-hard negatives. CAKE [77] introduces commonsense constraints into generating negatives. These strategies could improve the quality of negatives and reduce false negatives.

(e) **Entity Semantic Similarity-based Negative Sampling:** Replace entities with semantically similar ones to form harder, valid negatives. AN [132] uses K-means clustering. Truncated NS [133] exploits structural/attribute features. DNS [134], RCWC [135], and constraint-based [136] methods leverage types, contexts, and domain/range rules. LEMON [137] uses

TABLE I
POPULAR OPEN-SOURCE LIBRARIES FOR KGE

Library	Implementation	Key Features	GitHub Repository
OpenKE [144]	Pytorch, Tensor-Flow, C++	Efficiently implements fundamental operations such as data loading, negative sampling, and evaluation using C++.	https://github.com/thunlp/OpenKE
AmpliGraph [145]	TensorFlow	A Keras-style API with improved efficiency over OpenKE.	https://github.com/Accenture/AmpliGraph
torchKGE [146]	Pytorch	Efficiency: 2× over OpenKE and 5× over AmpliGraph.	https://github.com/torchkge-team/torchkge
LibKGE [147]	Pytorch	Enables direct configuration of hyperparameters and model settings via configuration files.	https://github.com/uma-pi1/kge
KB2E [34]	C++	The early KGE libraries and the predecessor of OpenKE.	https://github.com/thunlp/KB2E
scikit-kge [52]	Python	Implements multiple classical KGE models and supports a novel negative sampling strategy.	https://github.com/mnick/scikit-kge
NeuralKG [148]	Pytorch	Integrates KGE techniques with graph neural networks (GNNs) and rule-based reasoning models.	https://github.com/zjukg/NeuralKG
PyKEEN [149]	Pytorch	Offers 37 datasets, 40 KGE models, 15 loss functions, 6 regularization mechanisms, and 3 negative sampling strategies.	https://github.com/pykeen/pykeen
Pykg2vec [150]	Pytorch, Tensor-Flow	Supports automated hyperparameter tuning, and provides visualization for performance evaluation.	https://github.com/Sujit-O/pykg2vec
μKG [151]	Pytorch, Tensor-Flow	Supports multi-process execution and GPU-accelerated computation, making it well-suited for large-scale KGs.	https://github.com/nju-websoft/muKG
DGL-KE [152]	Pytorch, MXNet	Optimized for execution on CPU and GPU clusters, offering high scalability for large-scale KGs.	https://github.com/awsmlabs/dgl-ke
GraphVite [153]	Pytorch	Enables multi-processing and GPU parallelization to provide efficient large-scale embedding learning.	https://github.com/DeepGraphLearning/graphvite
PBG [154]	Pytorch	Designed for distributed training, capable of handling KGs with billions of entities and trillions of edges.	https://github.com/facebookresearch/PyTorch-BigGraph

PLM-derived descriptions. DHNS [138] integrates multimodal cues via diffusion models. However, introducing semantic relevance improves training but cannot fully avoid false negatives.

(f) **Efficient Negative Sampling:** NSCaching [139] caches high-quality negatives. MDNCaching [140] focuses on semantically invalid yet structurally similar negatives. TuckerDNCaching [141] uses Tucker factorization for better likelihood estimates. CCS [142] clusters entities to enhance cache utility. NS-KGE [143] reduces reliance on explicit negatives via a reformulated loss. These strategies are efficient but may undercut negative quality, limiting potential gains.

6) **Open-Source Library for KGE:** KGE has become a key research topic in the KGR field, leading to the emergence of numerous open-source libraries for development. We summarize several widely used and well-recognized KGE libraries on GitHub as in TABLE I, many surpassing 1000 stars.

Although KGE models have progressed rapidly, they map symbolic KGs to embeddings with limited interpretability, and often miss fine-grained semantic distinctions among candidate triples, which ultimately constrains performance.

B. Logic-based Model

Logic-based models leverage rules to provide precise and interpretable inference to retain the inherent explainability of symbolic KGs. A rule consists of a head atom implied by a conjunction of body atoms [155]. Horn rules, which contain only positive atoms, match KGs that typically store only positive facts. Given $Nationality(x, y) \Leftarrow$

$BornIn(x, z) \wedge LocatedIn(z, y)$, grounding with two observed triples ($David, BornIn, San\ Francisco$) and ($San\ Francisco, LocatedIn, U.S.A.$) yields a triple corresponding to the head atom ($David, Nationality, U.S.A.$).

To scale rule mining to large KGs, early work treats triples as binary predicates and applies inductive logic programming (ILP) [156] to learn first-order rules, including FOIL [157], MDIE [158], and Inspire [159], though scalability remains a critical limitation [160].

1) **Rule Learning for KG:** Rule learning approaches can be broadly categorized into two paradigms based on their construction strategy: inductive logic programming (symbolism) and neural network-based (connectionism) approaches.

(a) **Inductive Logic Programming:** For KGs, efficient rule learning models are designed, such as AMIE+ [161], ScaLeKB [162], and RDF2rules [163]. Among them, AMIE+ improves efficiency via optimized query execution and enhanced rule evaluation, while ScaLeKB introduces automatic data cleaning and partitioning. Most ILP-based models rely on exhaustive search and heuristic pruning, which can limit efficiency and rule quality. To address this, Evoda [164] introduces a genetic logic programming framework to extend Horn rules into generalized Datalog rules [165], incorporating a relational representation-based scoring scheme. Nevertheless, the computational demands of purely symbolic methods motivate hybrid, data-driven solutions for more scalable rule learning.

(b) **Neural Networks-based Model:** Neural networks-based approaches embed rule learning within differentiable frameworks. Both NeuralLP [166] and DRUM [155] leverage matrix

representations for relations and learn rule patterns in an end-to-end fashion. RLvLR [167] integrates KG embeddings and employs a customized scoring function to guide rule selection. RNNLogic [168] combines an RNN-based rule generator with a reasoning predictor, while RARL [169] separates rule generation and evaluation by modeling semantic relevance between rule components, enhancing scalability. Ruleformer [170] formulates rule mining as a sequence-to-sequence task, leveraging Transformer to encode subgraph context and predict relation probability distributions. Furthermore, Ott et al. [171] focus on rule aggregation, modeling dependencies among multiple rules to improve inference accuracy, addressing a key limitation in earlier rule-based methods.

2) **Joint Rule-KGE Model:** To effectively combine the explainability of symbolic rules with the generalization capabilities of KGE, two primary approaches have emerged: open-loop rule-enhanced KGE approaches and closed-loop iterative schemes that integrate rule learning with KGE.

(a) **Rule-Enhanced KGE:** Models such as KALE [172] and RUGE [173] incorporate symbolic rules into KGE to model relational patterns. For instance, the symmetric relation *hasFriend* leads to the rule $hasFriend(y, x) \Leftarrow hasFriend(x, y)$. These models instantiate rules with entities to generate new triples and apply t-norm fuzzy logic [174] to assess their truth values, enriching the KG and improving embedding quality, especially in sparse KG scenarios. However, these augmented triples may not always convey accurate semantics, and rules are often used only for data augmentation, not for guiding embedding directly. Rule [175] addresses this by jointly embedding entities, relations and rules, enabling confidence estimation for each rule to maintain consistency. However, converting symbolic rules into numeric forms would diminish explainability. Thus, RPJE [89] injects logical semantics into the embedding space via interpretable path composition, explicitly modeling relation associations, enhancing both accuracy and explainability during inference.

(b) **Iterative Integration of Rule Learning and KGE:** To more tightly couple rule learning and KGE, iterative frameworks like IterE [176], UniKER [177], and EngineKG [178] have been introduced. IterE extends KALE and RUGE by iteratively generating new triples from rules, updating the KG, and refining embeddings using t-norm fuzzy logic. UniKER follows a similar loop: rules infer new triples to expand the KG, which is then re-encoded via KGE, and embeddings are used for further reasoning. EngineKG modifies RPJE, integrating path-level semantic features with relation embeddings to accelerate rule learning and supporting a closed-loop system where rule mining and KGE reinforce each other. These iterative mechanisms continuously refine embeddings and uncover richer rules, thereby enhancing both reasoning performance and explainability.

3) **Comparison and Discussion:** TABLE II summarizes representative models for traditional KGR. Besides, we further give comparisons along five analytic axes:

(a) **Entity vs relation representation:** Ontology-aware and multimodal-enhanced KGE models focus on enhancing entity representation, while path-enhanced and rule learning models facilitate the intermediate relationships among entities and

emphasize logical association among relations. Basic KGE, negative sampling and joint rule-KGE paradigms provide equal treatment of entity and relation representations.

(b) **Training vs inference:** Training-centric components include negative sampling, ontology regularizers, path embedding, multimodal alignment, and rule-guided triple augmentation. Inference time arises from path aggregation, rule grounding, neuro-symbolic validation, or multimodal fusion.

(c) **Explainability vs generalization trade-off:** Basic KGE, negative sampling strategies and multimodal-enhanced schemes prioritize statistical coverage for generalization while offering limited explainability. Rule mining algorithms yield clear explainability but exhibit reduced generalization under noise and sparsity. Joint rule and KGE, path-enhanced KGE, ontology-aware KGE strike a middle ground by injecting explicit evidential or rules into embeddings, thereby preserving interpretable reasoning traces while mitigating data sparsity.

(d) **Scalability vs robustness:** Basic KGE models with advanced negative sampling ensure scalability. Multimodal-enhanced, ontology-aware, and path-enhanced models improve robustness under sparse or multi-source conditions. Joint rule-KGE models could balance scalability and robustness.

(e) **Data/resource requirements:** Ontology-aware models require high-quality ontological knowledge, path-based models need sufficient intermediate density, multimodal-enhanced approaches need aligned side information, rule miners need minimal relation support. Computational cost escalates from basic KGE models, then to path-enhanced models, logic-based models, and multimodal-enhanced models.

IV. MULTI-HOP KGR AND TEMPORAL KGR

In contrast to traditional KGR models that rely on direct relations between entities, multi-hop KGR models exploit paths that are naturally present in KGs, implying additional associations between entities and enhancing explainability [25]. Besides, temporal KGR models extend traditional KGR models to predict events by representing timestamps or modeling historical patterns under temporal constraints [26].

A. Multi-hop KGR Model

Multi-hop KGR infers the target (tail) entity by searching a path in a KG. Given the exponential growth of candidate paths, efficient exploration is crucial. Unlike path-enhanced KGE techniques aforementioned in Section III-A3, which pre-extract paths between entity pairs into embeddings, multi-hop KGR discovers informative paths at inference time, providing explicit, instance-level explanations. Beyond hard path search, soft reasoning paths model a distribution over multiple candidate paths to support KGR [179]. This section reviews two principal paradigms: reinforcement learning (RL)-based models and large language model (LLM)-based approaches.

1) **RL-based Model:** RL-based frameworks regard multi-hop KGR as a Markov Decision Process (MDP). Starting from the head entity, an agent sequentially selects relation-entity actions under a learned policy to reach the target entity, optimizing task rewards while requiring minimal supervision.

TABLE II
SUMMARIZATION OF REPRESENTATIVE MODELS FOR TRADITIONAL KGR.

Category	Main Idea	Representative Models	Strengths	Limitations
Basic KGE	Vector Translation Tensor Decomposition Neural Network GNN Transformer	TransE [32], TransH [33] RESCAL [12], ComplEx [51] NTN [28], ConvE [57] R-GCN [29], KBGAT [61] CoKE [67], SimKGC [103]	Simply and effectively encoding the topological features of KGs	Unexplainable, and performs poorly on small-scale KG
Ontology-Aware KGE	Ontology Graph Entity Type	JOIE [75], CAKE [77] TKRL [81], AutoETER [83]	Enriching high-level semantics	Ontological knowledge is limited
Path-Enhanced KGE	Relational Path Complete path Multiple paths	PTTransE [85], RPJE [89] RPE [91], Das et al. [92] PathCon [96], TAPR [98]	Enriching intermediate semantics with multi-step reasoning ability	Lacks precise path representation
Negative Sampling for KGE	Random uniform Probability-based Adversarial generation Self-supervised Entity semantic similarity Efficient Sampling	Random [33], Batch NS [120] Bernoulli [121], SparseNSG [123] KBGAN [124], GraphGAN [127] Self-Adv [47], CAKE [77] DNS [134], DHNS [138] NSCaching [139] NS-KGE [143]	Improving the quality of negative samples to promote training of KGE models	Difficult to ensure both high-quality samples and sampling efficiency
Multimodal-Enhanced KGE	Text-based Fusion-based Integration-based	DKRL [101], StAR [104] RSME [108], Native [113] MoSE [115], IMF [116]	Leveraging multimodal information to enhance entity representation	Noisy or poorly encoded modality might degrade overall robustness
Rule Learning	ILP Neural networks	AMIE+ [161], ScaLeKB [162] DRUM [155], Ruleformer [170]	Interpretable and accurate	Robustness is not as good as KGE models
Joint Rule-KGE Model	Rule-Enhanced KGE Iterative Mechanism	RUGE [173], RPJE [89] IterE [176], EngineKG [178]	Balancing explainability and robustness	With high complexity

(a) **Path search with deep RL:** DeepPath [25] first formulates multi-hop KGR with RL by treating relations as actions and rewarding successful arrivals at the tail entity, but it requires the assumption of given head and tail entities. MINERVA [180] removes this assumption by starting from the head entity and the query relation, encoding trajectory history with an LSTM and producing action probabilities via a feed-forward layer. DIVA [181] introduces variational inference to handle path uncertainty, adapting embeddings as the agent traverses ambiguous regions.

(b) **Reward shaping and policy enhancement:** To mitigate sparse binary rewards, MultiHopKG [182] uses soft rewards from pre-trained KGE models and action dropout to escape local optima. M-Walk [183] couples RNN-encoded paths with Monte Carlo Tree Search for better planning. RARL [184] injects high-quality logical rules to guide action selection, and AttnPath [185] enhances state encoding with attention-based LSTMs. DIVINE [186] employs generative adversarial imitation learning, where a discriminator reshapes rewards by distinguishing expert-like from random paths.

2) **LLM-based Model:** To improve generalization, recent works use LLMs as reasoning agents, leveraging rich pre-trained knowledge with little task-specific training.

(a) **In-context reasoning with prompts:** StructGPT [187] develops a call-linearize-generate loop: the LLM selects KG paths, verbalizes them, and generates answers. KSL [188] models retrieval as multi-step decisions to produce compositional, explainable paths. KD-CoT [189] runs a structured

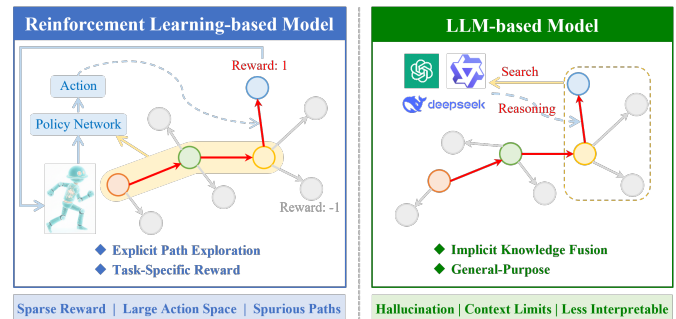


Fig. 5. The Comparison of RL-based and LLM-based Multi-hop KGR.

multi-turn QA loop that alternates LLM reasoning and retrieval. ToG [190] integrates beam search with LLM-based path ranking. KnowledgeNavigator [191] selects supporting evidence by modeling latent query constraints and building effective prompt templates. Nguyen et al. [192] propose discriminative and generative chain-of-thought (CoT) evaluation to score intermediate steps and final answers.

(b) **Instruction-tuned LLMs:** Prompt-only models might misalign language priors with KG topology, thus fine-tuning is required. KG-Agent [193] equips LLMs with tool-use and trains on structured KGR instructions. AgentTuning [194] builds a lightweight, mixed-domain instruction set to facilitate reasoning. GLaM [195] aligns LLMs to domain KGs by textualizing triples and pairing them with labeled instances, improving graph-grounded reasoning.

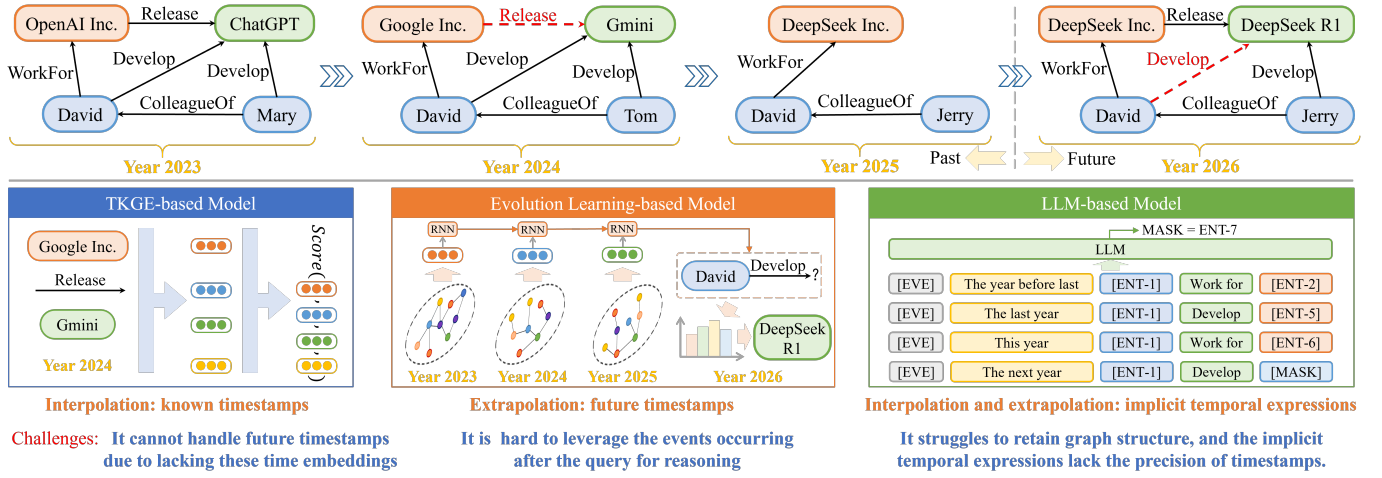


Fig. 6. The comparison of TKGE-based, evolution learning-based, and LLM-based TKGR models.

3) **Comparison and Discussion:** Figure 5 contrasts the two paradigms. An RL agent starts at the query’s subject entity and sequentially selects actions (namely relations) to traverse the KG, seeking a path that reaches the answer entity. Specifically, the agent’s policy network is trained to maximize the expected reward of finding a valid answer. Thus, the characteristics of RL-based models include: (a) **Explicit path exploration:** the learned agent selects actions step-by-step, yielding a concrete path as an explanation. (b) **Task-specific reward:** models require training on a given KG and well-designed rewards, learning policies tailored to that KG.

On the other hand, LLM-based models leverage pre-trained language models (such as GPT, Qwen, DeepSeek) to perform multi-hop KG reasoning, often by combining natural-language reasoning with KG access. The characteristics of LLM-based models include: (a) **Implicit knowledge fusion:** LLMs generate answers based on learned language patterns and implicit knowledge, which predict the final answer without exposing an internal path. (b) **General-purpose:** LLMs leverage vast pre-trained knowledge, enabling zero-shot or few-shot reasoning without extensive task-specific training.

B. Temporal KGR Model

In dynamic domains such as finance, news, and public security, data evolves continuously. Temporal KGs (TKGs) [196] extend static KGs by annotating facts with temporal scopes. Each fact is represented as an event quadruple $(s, p, o, time)$, where time denotes a timestamp or an interval indicating validity. Temporal KGR (TKGR) techniques can be grouped into three families: temporal KGE (TKGE)-based models, evolution learning-based models, and LLM-based models.

1) **TKGE-based Model:** TKGE-based models encode temporal information into embeddings and combine them with entity and relation embeddings to score event quadruples. TA-TransE [197] treats time as a numeric sequence and fuses it with relation embeddings using an LSTM, but lacks explicit time embeddings with limited precision. TTransE [198] incorporates timestamps into the TransE translation, yet struggles to distinguish temporally disjoint events with identical entities.

HyTE [199] projects entities and predicates onto time-specific hyperplanes inspired by TransH [33] to differentiate events across timestamps. TERO [200] replaces hyperplanes with RotatE-style rotations [47], improving cross-time alignment.

TDistMult [201] and TComplex [202] extend DistMult [50] and Complex [51] with temporal factors but assume time-invariant entity embeddings. DE-Simple [203] relaxes this by learning time-specific entity embeddings, while ATiSE [204] models temporal uncertainty via time-aware Gaussian distributions. TARGAT [205] combines temporal Transformers and graph attention to capture time-sensitive neighborhoods. LCGE [206] imposes temporal rules to regularize predicate embeddings for both timeliness and order. Thus, TKGE models are good at validity scoring but are limited on event ordering and long-range dependencies. ReMIT [207] leverages pre-training and knowledge distillation to fuse multi-level dynamics and yield stronger temporal embeddings.

2) **Evolution Learning-based Model:** These models construct sequences of KG snapshots (one subgraph per timestamp [23]), where GNNs encode local structure while sequence models such as LSTM/GRU capturing evolution. Know-Evolve [208] pioneers deep evolutionary KGs but cannot handle concurrent events. RE-NET [209] builds timestamped subgraphs and models them with GRUs, yet degrades when temporal continuity is weak. EvolveRGCN [210] updates GCN parameters across time for implicit adaptation. To capture recurrences, CyGNet [211] reuses past events. To reduce noise, CluSTeR [212] first retrieves history events via MDP and then reasons over it. Recent work further learns from relational correlations and periodic events to improve TKGR [213]. These models learn dynamics and causality, but often lack explicit timestamp representations and suffer from noisy histories or fragmented sequences.

3) **LLM-based TKGR Model:** LLM-based approaches textualize event quadruples and temporal context, and perform reasoning via pretrained LLMs. PPT [214] frames TKGR as masked word prediction with natural-language time cues (e.g., “the next day”, “one month later”). ECOLA [215] jointly encodes event structures and descriptions with multi-level

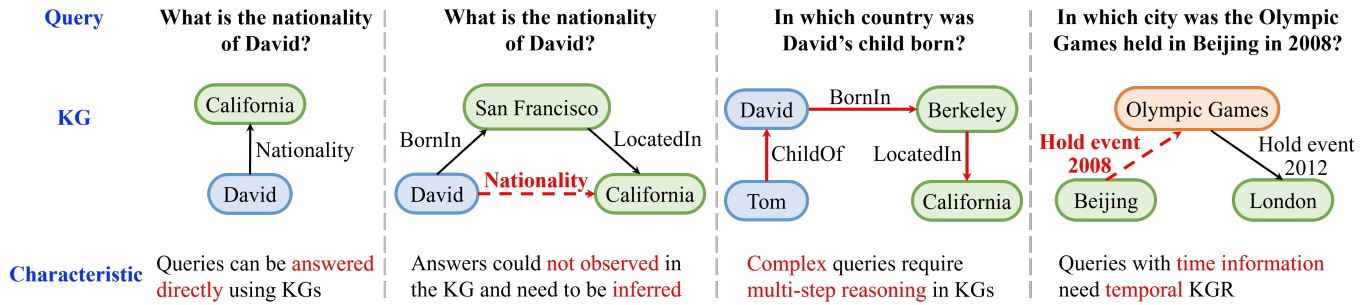


Fig. 7. Illustrative examples of the KGR technique applied to QA systems.

masking. SToKE [216] extends K-BERT [217] by constructing an Event Evolution Tree and injecting temporal/structural context via masked learning. GPT-NeoX [218] applies in-context learning over lexicalized event histories. For data augmentation, zrLLM [219] prompts GPT-3.5 to generate relational descriptions, aligns them with a T5 encoder, and fuses them with temporal KGE to improve low-resource prediction. LLM methods integrate rich semantics but may compromise graph-structural fidelity and often require careful prompting or alignment modules.

4) **Comparison and Discussion:** As shown in Fig. 6, we compare these TKGR streams along six axes:

(a) **Temporal representation:** TKGE learns explicit time embeddings, while evolution learning models capture dynamics via sequences of time-stamped snapshots. LLM-based models encode temporal context through textualized events and language priors.

(b) **Temporal granularity:** TKGE supports discrete points and continuous intervals. Evolution learning models operate on discrete snapshot steps. LLM-based models allow flexible, natural-language granularity.

(c) **Temporal reasoning focus:** TKGE emphasizes time-conditioned validity scoring. evolution learning models capture evolution and recurrence. LLMs perform temporally constrained reasoning with pre-trained knowledge.

(d) **Explainability:** TKGE and evolution learning models rely on dense embeddings and are less interpretable. LLMs can produce human-readable rationales.

(e) **Generalization:** TKGE may struggle with unseen entities and timestamps. Evolution learning models can generalize to new temporal patterns. LLMs exhibit zero-shot transfer to unseen graphs.

(f) **Suitable scenarios:** TKGE suits uniformly dense, validity-centric settings. Evolution learning models fit continuous streams with evolution/recurrence. LLMs handle implicit temporal expressions and cross-domain zero-shot tasks.

V. APPLICATION OF KGR

In real-world scenarios, KGR not only strengthens horizontal applications such as question answering (QA), recommendation, and visual reasoning, but also supports vertical domains including healthcare, commerce, cybersecurity, education, social governance, transportation, and environmental monitoring, thereby advancing knowledge-driven intelligent systems.

A. Downstream Tasks in Horizontal Domains

KGR has been widely applied in horizontal domains, including QA, recommendation, and visual reasoning by leveraging prior knowledge to improve accuracy and explainability.

1) **Question Answering Systems:** KG-based question answering (KGQA) is widely deployed in applications such as intelligent customer service and leverages structured KG information to deliver precise answers [220]. Nonetheless, KGQA presents three key challenges requiring advanced KGR: (a) inferring answers not explicitly present in the KG, (b) executing multi-hop reasoning for complex queries, and (c) addressing temporal constraints to obtain time-sensitive answers. Figure 7 overviews query types, their corresponding KGs, and characteristics of the QA process.

(a) **QA with Simple Queries:** To infer new knowledge and enhance QA robustness, KEQA [221] projects KGs into embedding space and formulates QA as a traditional KGR task aimed at predicting the tail entity with KGE models. TransE-QA [222] and TRL-KEQA [223] similarly integrate KGE into KGQA to infer previously unknown answers. These models effectively exploit latent KG features but are generally limited to single-hop queries.

(b) **QA with Complex Queries:** For multi-hop reasoning, EmbedKGQA [16] augments KEQA with BERT to encode complex questions and matches query and entity embeddings, but it neglects path-level information for effective multi-step reasoning. PKEEQA [224], PA-KGQA [225], and HamQA [226] explicitly inject path information into KGE, improving multi-hop inference. BRGNN [227] performs bidirectional path generation to capture richer semantics and improve accuracy. DSSAGN [228] fuses syntactic and semantic cues via an enhanced GNN to model the dependencies between the query and KG components for explainable multi-hop reasoning. Jiao et al. [229] incorporate relation embeddings to capture entity interactions, facilitating seamless traversal across multiple relations. CF-KGQA [230] further leverages causal structure to improve answer accuracy of multi-hop QA.

(c) **Temporal QA with Explicit Time Representations:** Temporal KGQA has recently garnered increasing attention as a significant application of temporal KGR techniques. TwiRGCN [231] applies a time-weighted GCN and a gating mechanism to decide whether the answer is a timestamp or an entity, which could guide scoring the candidates. CRONKGQA [18] scales EmbedKGQA [16] to temporal

queries by introducing a large temporal QA dataset and learning time-aware embeddings with temporal KGE. TempoQR [232] converts temporally annotated queries into entity- and time-aware representations and scores candidates via temporal KGE. However, such models struggle with implicit temporal expressions such as “before/after”, “first/next/last”.

(d) **Temporal QA with Implicit Temporal Constraints:** EXAQT [233] first extracts query-relevant dense subgraphs, learns time-aware entity embeddings via R-GCN [29], and reframes QA as node classification. Recent efforts combine LLMs with temporal KGQA to handle implicit constraints. Prog-TQA [234] introduces a self-evolving programming paradigm that exploits in-context learning to interpret temporal constraints. GenTKGQA [235] fuses temporally constrained subgraphs with LLM-generated textual representations and instruction-tunes open-source LLMs to better align temporal and structural semantics in retrieved events.

2) **Recommendation Systems:** In contrast to QA systems, recommendation systems are designed to generate personalized suggestions aligned with user preferences. To enhance recommendation quality, it is critical not only to provide recommendation results but also to enable deeper insights into user interests while ensuring explainability.

(a) **KGR for Mining User Preferences:** KGCN [236] is a pioneering model that employs GCNs to aggregate neighborhood information for encoding entities within a KG, effectively capturing both local and global structural semantics. The inherent reasoning capabilities of the learned entity embeddings contribute to a more profound understanding of item characteristics and user preferences. Extending this line of work, KGNCf-RRN [237] utilizes a modified residual recurrent neural network to encode relational paths between entities, thereby enabling multi-hop reasoning.

(b) **KGR for Enhancing Explainability of Recommendations:** To address the challenge of explainability, many existing models perform explicit reasoning over KGs by utilizing path-based information to evaluate the similarity between users or items [238]. KPRN [17] learns path embeddings by integrating entity and relation embeddings, and applies a weighted pooling mechanism to distinguish the relative importance of different paths, improving both performance and explainability. CogER [239] is the first to simulate dual-system human cognition in recommendation by employing an intuitive estimation phase (System 1) followed by a reinforcement learning-based reasoning phase (System 2), yielding high-quality recommendations along with interpretable reasoning paths. Furthermore, Lee et al. [240] combine KGs with textual review data by executing a GCN-based collaborative filtering algorithm over both the user-item interaction graph and the KG, thereby generating a rationale path from the target user to the recommended item. Finally, Markchom et al. [241] propose a meta-path grammar-based framework coupled with a machine translation model to transform complex meta-paths into natural language-like explanations.

(c) **Balancing Personalization and Explainability with KGR:** Many KG-based explainable recommendation models tend to ignore the heterogeneity of user preferences, leading to biased or unfair recommendations. To address this, Fu

et al. [242] introduce a fairness-aware model that integrates heuristic re-ranking based on multi-hop reasoning over KGs. KRRL [243] exploits a self-supervised reinforcement learning approach to enhance explainability in the domain of massive open online courses (MOOCs), leveraging both course content and user interest information encoded in a KG. Besides, several studies such as RippleNet [244], KGCN [237] and KGAT [245] integrate semantic information from KGs with multi-hop reasoning to improve the representation learning for users and items. These hybrid approaches combine the inherent transparency of path-based KGR models with a deeper exploration of user preference patterns, striking a balance between personalization and explainability.

3) **Visual Reasoning:** Recent advances in computer vision are shifting from low-level perception to higher-level cognition, motivating robust and explainable visual reasoning with KGR. We summarize three representative tasks: visual QA, cross-modal retrieval, and scene graph generation.

(a) **Visual Question Answering (VQA):** Knowledge-intensive VQA requires external or commonsense knowledge. FVQA [246] formalizes each instance as an image-question-answer-supporting-fact tuple to explicitly inject background knowledge. Wang et al. [247] link visual concepts to entities and exploit relational paths between query concepts and visual entities for reasoning. Graphhopper [248] first builds a scene graph and then uses RL-based multi-hop search to find answer paths. Hypergraph Transformer [249] constructs query-specific KGs by integrating question cues, visual targets, and external sources for reasoning. Recent studies [250] further show multimodal LLMs outperform traditional fine-tuned models on knowledge-intensive VQA.

(b) **Cross-Modal Retrieval (CMR):** The goal is to retrieve semantically aligned images/videos given text. KCR [251] extracts visual features via ResNet and enriches textual embeddings with external KGs using an enhanced BERT, optimizing alignment with a triplet loss. MMRG [252] builds image/text KGs and strengthens visual relational reasoning via pretraining tasks (e.g., attribute masking, contextual prediction). IRGR [253] jointly applies intra-modal, inter-modal, and instance-level KGR to capture relations across heterogeneous KGs and nearby instances, improving cross-modal similarity.

(c) **Scene Graph Generation (SGG):** SGG detects objects and predicts semantic relations, yielding a graph analogous to a KG. GB-Net [254] constructs initial scene graphs with Faster R-CNN, enriches them using ConceptNet [255] and WordNet, and refines node embeddings via message propagation while jointly reasoning over scene and external graphs. CGR [256] narrows the visual-knowledge semantic gap by composing knowledge routing paths through selective matching and retrieval, improving generalization. COACHER [257] mines neighbors and paths from commonsense KGs to enable robust, low-resource scene graph construction.

B. Downstream Tasks in Vertical Domains

In recent years, KGR has empowered applications across vertical domains, including healthcare, commerce, cybersecurity, education, social sciences, transportation, and environmental management.

1) **Healthcare Domain:** Zhu et al. [258] construct a disease-specific KG, namely SDKG-11, and employ multimodal KGR by projecting textual embeddings onto a hyperplane to uncover novel insights about specific diseases. Chai et al. [259] apply the KGE model TransE [32] to derive embeddings for a medical KG and then utilize a BiLSTM to predict associations between pathologies and diseases. Besides, both SSI-DDI [260] and KGNN [261] leverage KGs to capture higher-order structural and semantic relationships among drugs and relevant entities, enhancing the accuracy of drug-drug interaction predictions. SMR [262] integrates electronic medical records (EMR) with a medical KG, using KGE techniques to recommend safe medications. PharmKG [263] combines a heterogeneous graph attention network with global graph structures and heterogeneous features to capture complex biomedical information via learned embeddings. Moreover, KG-Predict [264] leverages KGE models to infer novel drug-disease interactions, demonstrating its effectiveness and practical value in drug repurposing for Alzheimer's disease.

In practice, healthcare deployments impose additional constraints beyond offline link prediction: (1) high cost of errors (false positives in medication recommendation can be harmful), (2) traceability and clinical plausibility of reasoning evidence for expert review, and (3) privacy and access control for patient records (such as EMR-linked KGs in SMR [262]). These requirements often favor models that can provide calibrated confidence and human-verifiable rationales, and that remain robust to noisy clinical facts.

2) **Business Domain:** Deng et al. [265] develop OpenBG, a multimodal commercial KG at a billion-scale, which is pre-trained by multimodal KGE models and vision-language models. This framework has been effectively applied to downstream e-commerce tasks such as product classification and prediction. Zhang et al. [266] propose a KGR approach based on graph attention networks, which generates transferable logical rules and is validated on real-world e-commerce KG datasets. KAPR [267] enhances reasoning accuracy via a multi-feature reasoning module that employs a dynamic strategy network for multi-hop reasoning over the KG, enabling predictions of product substitutability and complementarity. Mitropoulou et al. [268] apply various KGE techniques to discover latent connections among diverse customer needs and to identify implicit associations between customer requirements and product features. Kosasih et al. [269] integrate GNNs with neuro-symbolic machine learning techniques to infer multiple implicit relationships in supply chain risk management. Yang et al. [270] introduce temporal information to construct a dynamic KG for enterprise risk by aggregating neighbors and employing LSTM networks to model temporal patterns, achieving more accurate predictions of risk-related entities and relations. Besides, PKGM [271] employs a pre-trained KGE model on e-commerce product KGs, which is further applied to product recommendation and classification tasks.

Compared with benchmark settings, business systems are typically latency-constrained (online retrieval/ ranking), freshness-sensitive (rapidly changing products, prices, and user intent), and heterogeneous (multimodal content and inconsistent schemas as in OpenBG [265]). Thus, practitioners often

need to trade off accuracy with serving cost, and to handle distribution shifts and feedback loops during continuous deployment and A/B testing.

3) **Cybersecurity Domain:** Sikos [272] constructs a domain-specific KG for information security and employs rule learning-based reasoning models to automatically infer relations among vulnerabilities, weaknesses, platforms, and attack patterns. Qi et al. [273] utilize the rule learning algorithm AMIE [161] to mine logic rules and discover new network insights by learning rule patterns for each relation. Ezekia Gilliard et al. [274] apply a KGR technique based on the KGE model TransH to enhance the accuracy of automatic attack pattern recognition. Furthermore, to address the challenging issues of contract version diversity and limited detection time, SoliDetector [275] leverages a KG together with the solidity programming language and applies rule learning algorithms for reasoning, achieving efficient detection and localization of multiple defects. It can be discovered that rule learning-based KGR models are widely used in the cybersecurity domain due to the demand for explainability.

In real deployments, cybersecurity KGs must support near-real-time reasoning with rapidly evolving threats, and adversaries can deliberately exploit blind spots and data noise. Thus, besides accuracy, operational constraints include timeliness, robustness to incomplete/incorrect indicators, and actionable explanations for analysts, such as rule traces mined by AMIE [161], [273] or defect localization in SoliDetector [275].

4) **Other Domains:** In the field of education, Liang et al. [15] integrate GCNs with reinforcement learning to learn multi-path embeddings between entities, enabling efficient KGR and interpretable recommendations across diverse learning resources. In the domain of sociology, Zhou et al. [276] propose hierarchical KG embeddings that incorporate both global and domain-specific information, optimizing positional similarity loss within reasoning models to generate location-aware embeddings for the prediction of socio-economic indicators. In the tourism domain, Gao et al. [277] construct a tourism KG encompassing 340 cities and employ a GNN-based KGE model to deliver in-depth analyses of tourist preferences and attraction-level competition. In the transportation domain, SARGCN [278] pioneers the construction of a metro KG and assigns semantic types to each station, and then leverages a GNN to predict passenger flow. This model is validated using data from the subway systems of two Chinese cities, Shenzhen and Hangzhou. In the environmental science area, Liu et al. [279] first identify non-oil spill regions using predefined feature thresholds, and then enhance oil spill detection by combining rule-based learning with GNN-based KGR techniques, particularly in complex marine environments.

C. Summary and Discussion

KGR has been widely applied in horizontal domains such as QA, recommendation, and visual reasoning, as well as vertical domains including healthcare, commerce, cybersecurity, education, social sciences, transportation, and environmental management. In horizontal domains, KGR enhances accuracy and explainability by leveraging prior knowledge. In

vertical domains, KGR empowers domain-specific applications by capturing complex relationships and patterns. Across all applications, KGR has evolved from simple embedding-based models to sophisticated hybrid systems, increasingly integrated with LLMs. However, challenges remain in effectively integrating multimodal data, ensuring scalability to large KGs, and maintaining up-to-date knowledge in dynamic environments. Future research might focus on developing more robust and adaptable KGR techniques to address these challenges and further expand the KGR across diverse domains.

Furthermore, it is important to distinguish offline benchmark evaluation from real-world deployment. Benchmarks typically use fixed snapshots with curated train/ validation/test splits and report ranking metrics such as MRR and Hits@n under filtered settings (Section II). In contrast, deployed KGR systems are constrained by end-to-end requirements including (1) **latency** for online serving, (2) **data freshness and non-stationarity** as KGs evolve (new entities/relations, temporal drift), (3) **noise, privacy, and compliance** in domain data pipelines (such as EMR-linked healthcare KGs), and (4) **actionability** such as calibrated confidence and human-auditable rationales. Therefore, closing the benchmark-to-deployment gap often requires robustness, continual adaptation, and constraint-aware reasoning beyond optimizing offline link prediction scores.

VI. CHALLENGE AND OPPORTUNITY

While KGR has seen remarkable progress especially with neural and hybrid approaches, KGR approaches still show a clear performance gap in real deployments. Some challenges such as trustworthiness, multimodal reasoning, continual learning and uncertainty might be crucial to the next generation of KGR models.

A. Trustworthy Reasoning

As KGR techniques are increasingly deployed in high-stakes domains such as healthcare and cybersecurity, users require understandable justifications and confidence levels to validate inferred facts. Besides, many KGs implicitly or explicitly contain logical constraints, such as domain/range, symmetry/anti-symmetry, and transition. KGE models cannot guarantee satisfaction of such constraints, so inferred triples may violate logical rules. Traditional rule-based approaches inherently support explainability by offering explicit logical reasoning. Hybrid approaches such as IterE [176] and EngineKG [178] jointly learn logic rules and KG embeddings in a unified framework, combining symbolic explainability with neural scalability. Xu et al. [280] propose a human-centric evaluation platform for assessing the credibility of explanations in KGR. However, standardized metrics for explainability assessment are still lacking. To investigate the abductive reasoning on KGs, Zhang et al [281] introduce complex logical hypothesis generation to implement abductive KGR, proposing RLF-KG with reinforcement learning to enhance hypothesis generation by minimizing discrepancies between observations and conclusions. Thus, neuro-symbolic reasoning, unified metrics for explainability such as faithfulness with human-in-the-loop validation, and abductive reasoning might be critical

research frontiers on Trustworthy KGR. Furthermore, rules can be mutually inconsistent, requiring principled conflict resolution. Fuzzy logic relaxations trade strict correctness for differentiability, while global consistency is hard to enforce with local scoring and independent triple ranking. Therefore, future trustworthy KGR may benefit from unified designs that jointly address explainability and logical consistency.

B. Multimodal Reasoning

multimodal KGs integrate triples with multiple modalities such as text, images, and videos. Various multimodal KGR models focus on three main streams: (1) fusing KG embeddings with multimodal embeddings (such as MMKGR [114]), (2) negative sampling via multimodal semantics (such as DHNS [138]), and (3) leveraging LLMs to formulate the multimodal KGR as a classification task (such as VISITA [282]) or a generation task (such as MoMoK [283]). Future directions might include: (1) encoding models of conducting effective cross-modal alignment and fusion, and (2) generative models like the diffusion model of synthesizing missing modalities and enhancing robustness against noisy inputs.

C. Continual Learning

KGs constantly evolve as new information arrives while a KGR model should update without re-training from scratch. Thus, the challenge of continual learning on dynamic KGR involves avoiding catastrophic forgetting of old facts while incorporating new ones. LKGE [284] leverages a lifelong embedding learning strategy that utilizes a masked auto-encoder to update KG embeddings. Meanwhile, the transfer learning mechanism is employed to inject learned knowledge into the embeddings of new entities and relations. TIE [285] regards the deleted triples as negative triples and fine-tunes the model with new ones, effectively adapting to semantic changes in the KG. Although continual learning strategies can address issues related to parameter size, maintaining good reasoning performance and balancing between new and old knowledge remains an area worth exploring for future work.

D. Uncertain KGR

Uncertainty reasoning is fundamental in AI because real-world knowledge is inherently imprecise. Recent work on uncertain KGs augments facts with confidence scores, raising two core challenges: (1) representing degrees of uncertainty and (2) propagating / inferring them reliably. Beyond conventional point or tensor embeddings, distributional and geometric representations are explored: KG2E [39] models entities and relations as Gaussians with covariance encoding uncertainty. BEUrRE [286] uses axis-aligned rectangles with affine relational transformations. SUKE [287] exploits confidence-aware training to learn a confidence generator for inferred triples. MUKGE [288] introduces a resource-ranking based uncertainty reasoning algorithm. UKRM [289] treats rule mining as a seq2seq problem and applies differentiable reasoning (TensorLog [290]) before assigning confidences via a PLM. However, most approaches emphasize representation

while lacking unified mechanisms for uncertainty calibration, propagation, and evaluation during reasoning. Future directions include: (1) uncertainty-aware KGE frameworks that jointly model epistemic and aleatoric components, (2) probabilistic or interval-based multi-hop reasoning with principled uncertainty propagation, and (3) benchmark datasets annotated with calibrated confidence levels to standardize evaluation of uncertain KGR models.

E. LLM-enhanced KGR

LLMs bring strong semantic abstraction, controllable generation, and chain-of-thought (CoT) reasoning ability, motivating their integration with multiple KGR tasks [291].

1) **LLMs for Traditional KGR:** KG-GPT [292] retrieves task-relevant subgraphs and generates answers. MPIKGC [293] uses LLMs to (1) enrich entity descriptions, (2) paraphrase / interpret relations, and (3) summarize subgraph context for improved scoring. LARK [294] decomposes complex queries into KG context retrieval together with logical synthesis. ChatRule [295] prompts an LLM to induce symbolic rules from structural and textual cues. GENTKG [296] couples retrieval-augmented generation with temporal rule mining and lightweight instruction tuning for temporal facts.

2) **LLMs for Multi-hop KGR:** Works such as [187], [190], [192] probe CoT style path reasoning. Key open issues include: hallucinated steps, mismatch between generated chains and real KG paths, limited context window, and weak structural faithfulness [297]. Promising directions include: (1) neuro-symbolic alignment such as explicit path supervision and constrained decoding, (2) instruction tuning with domain knowledge for specific KGs, (3) tool augmentation KG retrievers such as rule miners.

3) **LLMs for Temporal KGR:** Beyond embedding timestamps or modeling snapshot evolution, LLMs enable: temporal abstraction (“the next week”, “subsequent quarter”), gap filling in incomplete event chains, and unification of heterogeneous granularities. Xia et al. [298] and Luo et al. [299] iteratively combine historical chain extraction with CoT style temporal inference (fine-tuning + generative reasoning). Future opportunities lie in jointly encoding (1) symbolic temporal constraints, (2) latent periodicity, and (3) commonsense progression within a controllable, uncertainty-aware generation framework.

4) **Limitations:** The role of LLMs in KGR also requires critical reflection on their limitations in deployment. **Hallucination and faithfulness** remain fundamental since LLMs may fabricate entities/relations or intermediate steps that are not supported by retrieved subgraphs, undermining reliability [297]. **Reasoning controllability** is non-trivial, since it is difficult to strictly constrain multi-step generation to valid KG operations under bounded computation budgets. **Consistency with symbolic knowledge** is another key challenge: generated facts may violate KG schema or logical constraints, and outputs can be unstable across prompts or runs. Therefore, beyond accuracy, future LLM-enhanced KGR should emphasize constraint-aware generation and verification such as logical rule-based checkers, provenance-aware retrieval, and evaluation that reports faithfulness/constraint-violation rates alongside task metrics [300].

In summary, future KGR systems are expected to evolve toward unified, uncertainty-aware, neuro-symbolic, multimodal, and continually adaptive reasoning frameworks powered but not dominated by LLMs, balancing scalability, interpretability, and reliability for deployment in actual applications.

VII. CONCLUSION

This survey systematically summarizes traditional KGR and multi-hop KGR as well as temporal KGR models. We also compare representative models to highlight their strengths and limitations. Furthermore, we explore emerging trends such as the integration of LLMs with KGR, and for the first time, propose a taxonomy of downstream applications from both horizontal and vertical perspectives. Finally, we outline key challenges, such as trustworthiness, multimodal reasoning, continual learning, uncertain reasoning and LLM-enhanced schemes to suggest future research directions aimed at bridging the gap between theory and practice. This work is intended to serve as both a reference and a roadmap for advancing KGR research area.

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Guanglin Niu received the bachelor's degree in automation from Beijing Institute of Technology in 2015, and the Ph.D. degree in computer science from Beihang University in 2022. He is currently a assistant professor with School of Artificial Intelligence, Beihang University. He has authored some top-tier conference papers, including AAAI, ACL, SIGIR, CVPR, EMNLP, ICME and COLING. His research interests include knowledge graph, natural language processing and multimodal learning.



Yangguang Lin received the bachelor's degree in computer science from School of Information Science and Technology, Beijing Forestry University in 2025. He is pursuing the master's degree in artificial intelligence at the School of Artificial Intelligence, Beijing University of Posts and Telecommunications. His primary research focus lies in the intrinsic tight coupling of knowledge graph and large language model.



Bo Li is currently a Changjiang Distinguished Professor with the School of Artificial Intelligence, Beihang University. He is a recipient of The National Science Fund for Distinguished Young Scholars. He is currently the associate dean of the School of Artificial Intelligence, Beihang University. He is the principal investigator of the National Key Research and Development Program. He has published over 100 papers in top journals and conferences and held over 50 domestic and foreign patents.